Learning Objective 2.4: I can describe how reflectors, Yagi-Uda antennas, and arrays achieve high gain, and select an appropriate high-gain antenna for a given application.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Sizing a C-band earth station.** A ground station uses a **3.0 m diameter** prime-focus parabolic reflector at **6.0 GHz**. Take the aperture efficiency to be $\eta_{\text{ap}} = 0.65$ and $c = 3 \times 10^8$ m/s.

   (a) In one or two sentences, explain why the statement "gain is fundamentally a statement about area" is more useful to a designer than "gain is how much the antenna amplifies."

   An antenna is passive: it amplifies nothing. What it does is collect the radiated power into a smaller solid angle, and the size of the coherent radiating area — measured in square wavelengths — is what sets how small that solid angle can be. Writing $G = \eta_{\text{ap}} 4\pi A/\lambda^2$ tells the designer the only two knobs that exist: make the aperture physically bigger, or use a shorter wavelength. "Amplification" suggests a third knob (more gain from the same box) that does not exist.

   (b) Find the wavelength, the dish diameter in wavelengths, and the gain in dBi.

   $G =$ $\boxed{43.6 \text{ dBi}}$

   $$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{6 \times 10^9} = 0.050 \text{ m} \quad \Rightarrow \quad \frac{D}{\lambda} = \frac{3.0}{0.050} = 60$$

   $$G = \eta_{\text{ap}} \left(\frac{\pi D}{\lambda}\right)^2 = 0.65\,(60\pi)^2 = 0.65\,(35531)$$

   $$= 2.31 \times 10^4 \;\rightarrow\; 10\log_{10}\left(2.31 \times 10^4\right) = 43.6 \text{ dBi}$$

   Remark: the effective aperture is $A_e = \eta_{\text{ap}} A = 0.65\,\pi(1.5)^2 = 4.60$ m^2 — that is the number Friis uses.

   (c) Estimate the half-power beamwidth, and state in one sentence what it implies about pointing.

   $\theta_{HP} =$ $\boxed{1.17^\circ}$

   $$\theta_{HP} \approx 70^\circ \frac{\lambda}{D} = 70^\circ \frac{0.050}{3.0} = 1.17^\circ$$

   A beam barely one degree wide means the mount has to hold the boresight to a small fraction of a degree; a few tenths of a degree of wind sway or thermal droop is already a measurable loss of signal. (Note also $2D^2/\lambda = 2(3)^2/0.05 = 360$ m — the far field starts a long way out, which is L9's problem.)

   (d) Management wants **3 dB more gain** at the same frequency. What dish diameter does that take, and name one specific cost of the change.

   Gain goes as D^2, so $+3$ dB is a factor of 2 in gain and a factor of $\sqrt{2}$ in diameter:

   $$D_{new} = \sqrt{2}\,(3.0) = 4.24 \text{ m} \qquad \theta_{HP,new} \approx 70^\circ \frac{0.050}{4.24} = 0.83^\circ$$

   The cost: the beam narrows to 0.83°, so the pointing budget tightens by the same factor of $\sqrt{2}$ — and the reflector, mount, and wind load all grow with D^2. (Accept also: surface tolerance must be held over twice the area, or the far field moves out to 720 m.)

2. **Where the efficiency went.** Same 3.0 m dish. The manufacturer quotes the following loss factors: spillover 0.88, illumination taper 0.86, blockage 0.96, and a lumped "everything else" factor of 0.98. The reflector surface is held to **0.5 mm RMS**. Ruze's formula gives the surface penalty as loss (dB) $= 685.8\,(\sigma/\lambda)^2$.

   (a) Spillover and illumination taper trade directly against each other. Explain the trade in one or two sentences, and state the rule of thumb for where the optimum sits.

   A narrow feed pattern puts almost all of its power on the dish, but leaves the rim many dB below the center, so the outer part of the aperture is not being used and the effective area shrinks. Widening the feed brightens the rim but sends power straight past it, which is lost entirely (and on receive replaces cold sky with 290 K ground in the spilled beam). The optimum is the familiar rule of thumb: illuminate the **rim about 10 dB below the center**.

   (b) Compute the aperture efficiency from the four quoted factors, before any surface error. Call it $\varepsilon_{ap,1}$.

   $\varepsilon_{ap,1}$ = $\boxed{0.712}$

   $$\begin{aligned}\varepsilon_{ap,1} &= (0.88)(0.86)(0.96)(0.98)\\ &= 0.712\end{aligned}$$

   (c) Add the Ruze surface term at 6.0 GHz and report the total aperture efficiency.

   η_{ap} = $\boxed{0.70}$

   At 6 GHz, $\lambda = 50$ mm, so $\sigma/\lambda = 0.5/50 = 0.010$.

   $$\begin{aligned}\text{loss} &= 685.8\,(0.010)^2 = 0.069\text{ dB}\\ \eta_{\text{surf}} &= 10^{-0.069/10} = 0.984\\ \eta_{\text{ap}} &= (0.712)(0.984) = 0.700\end{aligned}$$

   A half-millimetre surface is essentially free at C band.

   (d) Someone proposes reusing this *same physical dish* at **30 GHz**. Recompute the surface term and the total aperture efficiency, then say in one or two sentences what the result means for the proposal.

   At 30 GHz, $\lambda = 10$ mm, so $\sigma/\lambda = 0.050$:

   $$\begin{aligned}\text{loss} &= 685.8\,(0.050)^2 = 1.71\text{ dB}\\ \eta_{\text{surf}} &= 10^{-0.171} = 0.674\\ \eta_{\text{ap}} &= (0.712)(0.674) = 0.48\end{aligned}$$

   Nothing about the dish changed — only λ — and the aperture efficiency fell from 0.70 to 0.48. The dish still gains about 56.3 dBi at 30 GHz (versus 58.0 dBi with a perfect surface), so the proposal is not absurd, but 1.7 dB has been thrown away for free and it gets exponentially worse with frequency. Above roughly 30 GHz this reflector is a machining problem, not an antenna.

3. **Reading a Yagi-Uda.** A Yagi-Uda has one driven element, one reflector behind it, and a row of directors in front along a boom.

   (a) The reflector is cut slightly *longer* than resonance and the directors slightly *shorter*. Explain what that detuning does electrically, and how it produces an endfire beam.

   Only the driven element is connected; the parasites carry currents *induced* by its near field, and the phase of that induced current is set by the element's reactance. A slightly long element is inductive, so its current lags; a slightly short element is capacitive, so its current leads. Placing the lagging element behind and the leading elements in front makes the re-radiated fields add in phase along the boom in the forward direction and tend to cancel in the reverse direction. The result is an endfire main lobe pointing away from the reflector, with a front-to-back ratio of roughly 15 to 25 dB.

   (b) A second reflector buys almost nothing, but a seventh director still helps. Why?

   The first reflector already cancels most of the field behind the antenna, so a second one sits in a region where there is almost no field to couple to and almost nothing left to cancel. The directors, by contrast, sit in the forward beam where the field is strong, and each one extends the slow forward traveling wave over a longer aperture. Gain follows the length of that aperture — the boom — not the number of rods on it.

   (c) A 6-element Yagi on a 1.0λ boom gives 10 dBi. Using the rule of thumb of $+3$ dB per doubling of boom length, estimate the boom needed for 16 dBi at 435 MHz. Then say whether you believe the answer.

   L = $\boxed{2.76\text{ m}}$

   $16 - 10 = 6$ dB, which is two doublings, so the rule of thumb predicts a 4λ boom. At 435 MHz, $\lambda = 3\times10^8/435\times10^6 = 0.69$ m:

   $$L = 4\lambda = 4\,(0.69) = 2.76\text{ m}$$

   Do not believe it. The rule of thumb is optimistic past a few wavelengths: published designs put a 4.5λ boom at about 14.5 dBi, not 16. Realistically you would either accept a much longer boom with diminishing returns, or stack two 13 dBi Yagis and take the $+3$ dB from the stack.

   (d) Yagis are narrowband — a few percent. Explain why, and name one application where that does not matter.

   Every element is a detuned resonator, and the whole design depends on the *phases* of the induced parasite currents. Move off the design frequency and each element's reactance changes, the phases drift, and the forward addition and rearward cancellation both degrade — gain drops and the front-to-back ratio collapses. That is fine anywhere the frequency is fixed: a single-channel broadcast TV antenna, an amateur band antenna, or a fixed point-to-point link.

4. **Choosing an antenna.** You need a **25 dBi** antenna at each end of a 2 km rooftop-to-rooftop link at **5.8 GHz**. The mounts are fixed (no steering), the site is windy, and the budget is small.

   (a) Find the required effective aperture, then the diameter of a parabolic dish that delivers it with $\eta_{\text{ap}} = 0.60$.

   D = 0.38 m

   $\lambda = 3 \times 10^8 / 5.8 \times 10^9 = 0.0517$ m and $G = 10^{25/10} = 316$.

$$A_e = \frac{G\lambda^2}{4\pi} = \frac{316\,(0.0517)^2}{4\pi} = 0.0673\ \text{m}^2$$
$$A = \frac{A_e}{\eta_{\text{ap}}} = \frac{0.0673}{0.60} = 0.112\ \text{m}^2$$
$$D = 2\sqrt{A/\pi} = 2\sqrt{0.0357} = 0.378\ \text{m}$$

   Beamwidth: $\theta_{HP} \approx 70^\circ(0.0517/0.378) = 9.6^\circ$.

   (b) A vendor offers a flat patch array instead, with $\eta_{\text{ap}} = 0.75$ and elements on a $\lambda/2$ grid. How many elements, and how big is the panel?

   N = 144

$$A = \frac{A_e}{0.75} = \frac{0.0673}{0.75} = 0.0898\ \text{m}^2 \quad \Rightarrow \quad \text{side} = 0.300\ \text{m}$$
$$\frac{\lambda}{2} = 0.0259\ \text{m} \quad \Rightarrow \quad \frac{0.300}{0.0259} = 11.6 \rightarrow 12$$

   A $12 \times 12 = 144$ element panel, about 0.31 m square. Check: $G = 0.75\,(4\pi)(0.31)^2/(0.0517)^2 = 339 = 25.3$ dBi . Good.

   (c) Pick one and defend it in two or three sentences, using the numbers you just computed.

   Take the dish. It is a single moulded part 0.38 m across with one feed, it is broadband, and its wind load at that size is modest; the patch array needs 144 elements and a 144-way corporate feed network whose loss at 5.8 GHz is significant in both cost and dB. The array's advantages — flat profile and the option of electronic steering — are worth nothing on a fixed rooftop link, so they do not justify the cost. (A defensible opposite answer: choose the panel if the roof lease or the wind survey forbids a dish, and accept the feed loss.)

   (d) The dish beam is 9.6° wide. Using the beam-shape approximation $L(\theta) \approx 3\,(2\theta/\theta_{HP})^2$ dB, how far off boresight can the mount drift before you lose 0.5 dB?

   θ = $\pm 2.0^\circ$

$$0.5 = 3\left(\frac{2\theta}{9.6^\circ}\right)^2 \quad \Rightarrow \quad \frac{2\theta}{9.6^\circ} = \sqrt{0.1667} = 0.408$$
$$\theta = \frac{(0.408)(9.6^\circ)}{2} = 1.96^\circ$$

   About $\pm 2^\circ$ — comfortable for a rigid rooftop mount, which is another argument for not buying more gain than the link needs.

5. **Does the link close?** Continue the 5.8 GHz, 2 km link from the previous question. The transmitter delivers **20 dBm** and the receiver sensitivity is -80 **dBm**. Use Friis, $P_r = P_t G_t G_r (\lambda/4\pi R)^2$.

   (a) Compute the free-space path loss in dB.

$$L_{fs} = 20\log_{10}\left(\frac{4\pi R}{\lambda}\right) = 20\log_{10}\left(\frac{4\pi(2000)}{0.0517}\right)$$
$$= 20\log_{10}\left(4.86\times10^5\right) = 113.7\text{ dB}$$

L_{fs} = $\boxed{113.7\text{ dB}}$

   (b) Find the received power with the 25 dBi dishes at both ends, and the link margin.

P_r = $\boxed{-43.7\text{ dBm}}$

$$P_r = P_t + G_t + G_r - L_{fs}$$
$$= 20 + 25 + 25 - 113.7 = -43.7\text{ dBm}$$
$$\text{margin} = -43.7 - (-80) = 36.3\text{ dB}$$

   (c) Now replace both dishes with 8 dBi patches. Recompute P_r and the margin.

P_r = $\boxed{-77.7\text{ dBm}}$

$$P_r = 20 + 8 + 8 - 113.7 = -77.7\text{ dBm}$$
$$\text{margin} = -77.7 - (-80) = 2.3\text{ dB}$$

The 34 dB of antenna gain you gave up came straight out of the margin.

   (d) Which link would you actually deploy, and why? Name two specific things that eat margin on a real rooftop link.

Deploy the dish link. A 2.3 dB margin leaves no room for normal propagation variation, so the link fails on the first bad day. Real margin eaters include rain and wet foliage in the path, mount drift and building sway pushing the beam off boresight (Q4d showed only $\pm 2^\circ$ of slack), multipath from nearby rooftops, and connector and cable loss between the radio and the antenna. With 36 dB in hand the dish link absorbs all of that; with 2.3 dB it absorbs none of it.

Documentation: ____